

AI Customer Support Agent for @AppleSupport

Hiver SDE Intern Take-Home Technical Report | Evaluation-Focused AI Prototype

Executive Summary

Social media customer support on Twitter/X presents acute operational constraints: interactions are public, real-time, and capped at 280 characters. For a premier consumer brand like Apple, an AI agent must not only provide high-accuracy technical guidance; it must know with statistical reliability **when not to answer**. This report presents an evaluation-focused AI Support Agent prototype for @AppleSupport, benchmarked on a 200-sample hand-audited, 100% genuine holdout Golden Evaluation Set. The holdout gold set is strictly thread-disjoint from both the 600 training threads and the 1,000 historical KB retrieval corpus (with training threads contained within the KB). Across this benchmark, the proposed agent achieves: **62.5% Out-of-Sample Intent Accuracy** across a 7-class domain taxonomy (vs. 36.0% Trivial, 55.0% Simple Baseline), **69.2% Escalation Recall** on safety-critical interactions (vs. **0.0%** for Trivial and **3.9%** for Simple Baseline), **79.5% Official Apple Domain Link Validity**, and **100% Twitter character compliance**. The full evaluation harness reproduces deterministically on standard CPU in **8.2 seconds**.

1. Problem Framing & Scope Boundaries

Operational Principles for @AppleSupport:

- **Safety and Privacy First:** Public social threads must never handle account credentials, passwords, 2FA codes, or billing card details. The agent immediately escalates to secure DM routing and authenticated recovery portals.
- **Definitive Actionability:** Concrete, verifiable UI paths (e.g. *Settings > Battery > Battery Health*) accompanied by verified canonical Apple URLs (support.apple.com/HT...).
- **Apple Brand Persona:** Calm, professional, empathetic tone within Twitter's 280-character limit.

Explicit Non-Goals:

- *No Automated Financial Payouts:* All refund and subscription cancellation requests require authenticated human review to prevent social media bot fraud.
- *No In-Chat Credential Resets:* Account lockouts route exclusively to iforgot.apple.com.
- *No Machine-Translated Technical Troubleshooting:* Non-English queries are routed to native-language queues to prevent dangerous mistranslation of recovery procedures.
- *No Speculative Hardware Cost Estimation:* Physical component failures route to official Genius Bar scheduling.

2. Empirical Results vs. Two Baselines

Comparative performance on N=200 Golden Evaluation Set (Thread-Disjoint Holdout Split):

Metric Dimension	Trivial Baseline (Always Auto-Handle)	Simple Baseline (Naive Bayes + 1-NN)	Proposed AI Agent (RAG + Policy Engine)
Intent Accuracy (Out-of-Sample)	36.0%	55.0%	62.5%
Intent Macro F1	0.076	0.267	0.474
Brier Calibration Score (Lower=Better)	1.0	0.7874	0.6023
Expected Calibration Error (ECE)	0.36	0.355	0.054
Escalation Accuracy	74.0%	75.0%	71.0%
Escalation Recall (Safety-Critical)	0.0%	3.9%	69.2%
Escalation Precision	0.0%	100.0%	46.2%
Escalation F2 Score (Recall-Weighted)	0.000	0.048	0.629
False Escalation Rate (Lower=Better)	0.0%	0.0%	28.4%
Illustrative 5:1 Risk Penalty (5*FN + 1*FP)	260	250	122
SacreBLEU Score	0.2	0.4	4.1
Twitter Char Limit Compliance (<280)	100.0%	95.0%	100.0%
Official Domain Inclusion Rate	0.0%	0.0%	79.5%
Intent-Link Relevance Rate	0.0%	0.0%	75.8%

Heuristic: Groundedness (1-5)	4.20	4.06	4.78
Heuristic: Brand Voice & Empathy (1-5)	5.00	4.36	4.58
Heuristic: Actionability (1-5)	4.20	4.33	4.80
Heuristic: Escalation Appropriateness	4.13	4.17	4.42
Heuristic: Overall Quality Score (1-5)	4.38	4.23	4.65

The Illusion of Baseline Accuracy: The Trivial Baseline achieves 74.0% accuracy simply by auto-handling all inquiries. However, its **Escalation Recall is 0.0%**, missing every safety-critical ticket. The Proposed Agent catches **69.2% of escalations** (36 of 52), reducing the illustrative 5:1 weighted risk penalty from 260 to 122.

3. Failure Mode Analysis (Top 5 Real Observed Failures)

Failure Mode 1: Sarcasm and Ambiguous Frustration [GOLD_002]

- *Query:* "It's been nearly two weeks and I've yet to get LTE on my Watch () working. Think I paid a premium for a spec of red paint."
- *Observed Output:* Predicted Intent = OUT_OF_SCOPE_OTHER (Conf: 0.49). Decision = AUTO_HANDLE.
- *Gold:* Intent = CUSTOMER_FEEDBACK_COMPLAINT, Escalation = ESCALATE.
- *Root Cause & Mitigation:* Sarcastic complaint lacked explicit vulgarity or legal threats, and confidence (0.49) was above the low-confidence threshold (0.40). Mitigation: Integrate multi-week duration and churn phrase detectors.

Failure Mode 2: Multilingual Language Routing Miss [GOLD_004]

- *Query:* "Mira que me gusta vuestra actualización pero me va como el culo ahora, cuando queráis lo solucionáis."
- *Observed Output:* Predicted Intent = OUT_OF_SCOPE_OTHER (Conf: 0.69), Decision = AUTO_HANDLE.
- *Gold:* Intent = OUT_OF_SCOPE_OTHER, Escalation = ESCALATE (Non-English routing).
- *Root Cause & Mitigation:* Idiomatic Spanish ('como el culo') evaded the localization regex, auto-handling in English. Mitigation: Pre-classification language identification filter (fastText/langdetect).

Failure Mode 3: False Escalation on Repeated Troubleshooting Phrasing [GOLD_008]

- *Query:* "I've had to do it multiple times when I reset my device and how do I change trusted device from old phone to new phone?"
- *Observed Output:* Decision = ESCALATE (via POLICY_REPEATED_UNRESOLVED_FAILURE).
- *Gold:* Decision = AUTO_HANDLE (Routine device pairing how-to).
- *Root Cause & Mitigation:* The phrase 'multiple times' triggered the repeated failure heuristic on a standard UI inquiry. Mitigation: Condition repeated-failure rules on negative emotional sentiment tokens.

Failure Mode 4: Low-Confidence Safety Interception [GOLD_019]

- *Query:* "Cool new feature in macOS High Sierra, it knows you've been working too hard and freezes the screen, but not the mouse."
- *Observed Output:* Intent = IOS_SOFTWARE_UPDATE (Conf: 0.37). Escalation = ESCALATE (via POLICY_LOW_MODEL_CONFIDENCE).
- *Gold:* Intent = IOS_SOFTWARE_UPDATE, Decision = AUTO_HANDLE.
- *Root Cause & Mitigation:* Low classifier confidence (< 0.40) safely prevented automated troubleshooting on a sarcastic OS freeze complaint, routing to human support.

Failure Mode 5: Sensitive Domain Precautionary Escalation [GOLD_022]

- *Query:* "Grrr! Why won't the contacts from my phone back up to iCloud?? #icloud"
- *Observed Output:* Intent = APPLE_ID_AND_ICLOUD (Conf: 0.77). Escalation = ESCALATE (via POLICY_SENSITIVE_DOMAIN).
- *Gold:* Intent = APPLE_ID_AND_ICLOUD, Escalation = AUTO_HANDLE.
- *Root Cause & Mitigation:* Broad sensitive domain rules escalate all Apple ID/iCloud inquiries to safeguard account privacy, causing false escalation on routine contact syncing.

4. 'What is Misleading About My Headline Number?'

A responsible engineering evaluation must transparently acknowledge the limitations of offline benchmark figures:

1. **Single-Turn Static vs. Multi-Turn Dynamic Evaluation:** Our benchmark evaluates the incoming customer message in isolation. In production, customer interactions span multiple back-and-forth turns.
2. **Illustrative Asymmetric Cost:** Under an illustrative 5:1 penalty weighting missed escalations more heavily than unnecessary escalations, the policy reduces the penalty from 260 to 122. However, this is a candidate-selected metric, not an empirical dollar figure.
3. **Domain Taxonomy Conditioning:** Achieving 62.5% out-of-sample intent accuracy reflects a closed 7-class taxonomy. In an unconstrained open-vocabulary setting, intent accuracy would degrade.
4. **69.2% Escalation Recall in Production:** Catching 36 of 52 escalations means 16 are missed. In production, we would not deploy unrestricted auto-reply at this threshold; initial rollout must operate as an agent-assist copilot.

5. Candidate Human Annotator vs. LLM-as-a-Judge Agreement (N=50)

To assess automated grading reliability, we conducted an inter-rater agreement study comparing Gemini 2.5 Flash rubric scoring and candidate author blind scoring on the exact same 50 frozen agent outputs: **Pearson Correlation $r = 0.934$, Spearman $\rho = 0.735$, Mean Absolute Error = 0.237 points** (on 1–5 scale), with **100.0% of ratings within 0.5 points**. SHA256 validation ensures both rating files refer to the same frozen candidate outputs and prevents accidental reuse of ratings from changed model responses.

6. What We Would Do Next with One More Week

1. **Tri-State Copilot Routing:** High confidence (>0.85) -> automated reply; Medium confidence ($0.55-0.85$) -> draft in Hiver inbox for one-click human agent approval; Low confidence (<0.55) -> direct human routing.
2. **Stateful Multi-Turn Conversation Memory:** Track user dialogue history across multiple tweets to disambiguate intermittent hardware vs. software symptoms.
3. **Mock CRM Tool Calling:** Query Apple System Status API and serial-number warranty entitlement endpoints prior to drafting responses.